FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2016**

HALF COURSE TEST – VIII

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section – A (01 – 06)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

   Section – A (07 – 09) contains 3 multiple choice questions which have one or more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

   Section-A (10 – 14) contains 2 paragraphs. Based upon paragraph, 2 and 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.
2. **Section – B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct matching. No marks will be given for any wrong matching in any question. There is no negative marking.
3. **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. A tank is filled with water to a height H. Two holes are made on its side wall, one at height of h from the bottom and the other at a depth h from the top. The horizontal jets starting from the two holes meets outside the tank at some point. The distance of this point from the side of the tank is

(A) $\sqrt{[4h(H-h)]}$ (B) $\sqrt{[4H(H-h)]}$

(C) $\sqrt{[2h(H-h)]}$ (D) $\sqrt{[3h(H-h)]}$

2. Shampoo is being poured into a vessel of frustum shaped with constant volume flow rate 16.5 cm^3/s. At what rate is its level in the bottle rising at a point where diameter of the bottle is 6.30 cm. (keeping significant figures in view)

(A) 0.063 cm/s (B) 0.63/cm/s
(C) 0.529 cm/s (D) 0.5291 cm/s

3. A particle moves along x-axis following the relation $px^2 + qv^2 = r$, where p, q, r are positive constants. x is co-ordinate of the particle and v is the instantaneous speed. The time interval between two consecutive instants, when particle is at rest is

(A) $2\pi\sqrt{\frac{q}{p}}$ (B) $\pi\sqrt{\frac{q}{p}}$

(C) $\pi\sqrt{\frac{p}{q}}$ (D) $2\pi\sqrt{\frac{p}{q}}$

Space for rough work

4. A certain quaternary star system consist of three star, each of mass 'm' moving in same circular orbit of radius 'r' about a central star of mass 'M'. The stars orbit in same sense and are positioned one-third of circumference apart from one another. The period of revolution of each star will be

(A) $2\pi\sqrt{\frac{r^3}{G\left(M+\frac{m}{\sqrt{3}}\right)}}$ (B) $2\pi\sqrt{\frac{r^3}{G(M+m)}}$

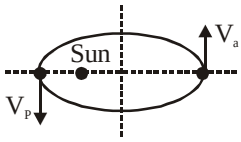
(C) $2\pi\sqrt{\frac{r^3}{G(M+3m)}}$ (D) $2\pi\sqrt{\frac{r^3}{GM}}$

5. If velocity of sound in Hydrogen at certain temperature is v_0. The velocity of sound in a mixture of two parts by volume of Hydrogen and one part of oxygen at same temperature is

(A) $v_0\sqrt{6}$ (B) $v_0\sqrt{3}$

(C) $\frac{v_0}{\sqrt{6}}$ (D) $\frac{v_0}{\sqrt{3}}$

6. A planet moves around Sun in an elliptical orbit of eccentricity e. The ratio of the velocity at perigee V_p and at apogee V_a is given by



(A) $\frac{V_P}{V_a}=\frac{1+e}{1-e}$ (B) $\frac{V_P}{V_a}=\frac{1-e}{1+e}$

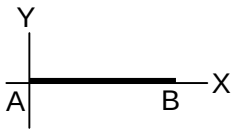
(C) $\frac{V_P}{V_a}=\sqrt{\frac{1+e}{1-e}}$ (D) $\frac{V_P}{V_a}=\sqrt{\frac{1-e}{1+e}}$

Space for rough work

Multiple Correct Choice Type

This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. A wire of mass M and length L is placed along x-axis with end A at origin. The linear mass density λ (of the rod) varies as $\lambda = \lambda_0 x$, where x is distance from end A, λ_0 is a constant.



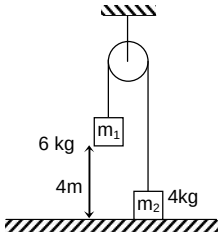
(A) The value of λ_0 is $\frac{2M}{L^2}$

(B) Moment of Inertia of the rod about an axis passing through A and parallel to y-axis is $\frac{1}{2}ML^2$

(C) Moment of inertia of the rod about an axis passing through COM of the rod and parallel to y-axis is $\frac{ML^2}{18}$.

(D) Moment of inertia of the rod about an axis passing through B and parallel to y-axis is $\frac{1}{6}ML^2$.

8. Two objects are connected by a light string passing over a light frictionless pulley as shown. The object of mass 6 kg is released from rest. ($g = 10\ m/s^2$)



(A) Time taken by object of mass 6 kg to strike the surface is 2 sec.
(B) Maximum height attained by block of mass 4 kg is 4.8 m.
(C) Duration after which 6 kg block will leave the surface for the first time is 2.8 sec.
(D) None of the above.

9. A boy in an elevator with open roof shoots a bullet in vertically upward direction from a height of 1.5 m above the floor of the elevator. The initial speed of the bullet with respect to elevator is 15 m/s. The bullet strikes the floor after 2 seconds. Then (Assuming $g = 10\ m/s^2$)
(A) Lift is moving with constant speed
(B) Lift is moving with upward acceleration of $5.75\ m/s^2$
(C) Lift is moving with downward acceleration of $5.75\ m/s^2$
(D) Distance travelled by bullet during its flight can not be calculated from the given data.

Space for rough work

Paragraph Type

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple** choice questions and based on the other paragraph **3 multiple** choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Questions Nos. 10 and 11

In a standing wave experiment a 1.2kg horizontal rope is fixed in place at its two ends ($x = 0$ and $x = 2.0m$) and made to oscillate up and down in the fundamental mode, at frequency 5.0Hz. At $t = 0$ the point at $x = 1.0m$ has zero displacement and is moving upward in the positive direction of y-axis with a transverse velocity $3.14ms^{-1}$.

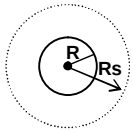
10. Amplitude of the motion of the particle at $x = 1.0m$ is
(A) 0.16m (B) 0.1m
(C) 0.08m (D) 0.05m

11. Displacement of the particle at $x = 0.5m$ and time $t = 0.15\,\text{sec}$ is
(A) $\frac{1}{5\sqrt{2}}m$ (B) $-\frac{1}{5\sqrt{2}}m$
(C) $-\frac{1}{10\sqrt{2}}m$ (D) $-\frac{1}{20\sqrt{2}}m$

Space for rough work

Paragraph for Questions Nos. 12 and 14

Supernova refers to the explosion of a massive star. The material in the central case of such a star continues to collapse under its own gravitational pull. If mass of the core is less than 1.4 times the mass of sun, its collapse finally results in a white dwarf star. However, if the core has a mass greater than this, it could end up soon as a neutron star and if its mass is more than about three solar masses, the collapse may still continue till the star becomes a very small object with an extremely high value of density called a 'Black hole'. Escape speed for a black hole is very large. The figure shows a block hole of radius R and another concentric sphere of radius R_S, called the 'Schwarzschild Radius'. It is the critical radius at which escape speed equals the speed of light c. Nothing even the light, can escape from within the sphere of Radius R_S. So light from a black hole cannot escape and hence the terminology 'black hole'.



There has been astronomical evidence of a small and massive object at the centre of our galaxy the 'Milky way'. Suppose that there is a particle at a distance about 6 light years that orbits this massive object with an orbital speed of about 2×10^5 m/s. Use the given data wherever necessary and answer the questions that follow.

$G=6.67\times10^{-11}$ N–m^2/kg^2, Solar mass $M=2 \times 10^{30}$ kg, $C = 3 \times 10^8$ m/s, 1 light year = 9.5×10^{15} m.

12. Mass (in kg) of the massive object at the centre of the milky way galaxy is of the order.
(A) 10^{32} (B) 10^{37}
(C) 10^{43} (D) 10^{29}

13. Theories suggest that it is not possible for a single star to have a mass of more than 50 solar masses. The massive object at the centre of milky way galaxy is most likely to be a
(A) white dwarf (B) neutron star
(C) black hole (D) single ordinary star

14. If mass of earth $M_E \approx 6\times10^{24}$ kg and its radius R_E = 6400 km, to what fraction of its presents radius does the earth need to be compressed in order to become a black hole? (give only the order of your answer)
(A) 10^{-4} (B) 10^{-9}
(C) 10^{-7} (D) 10^{-14}

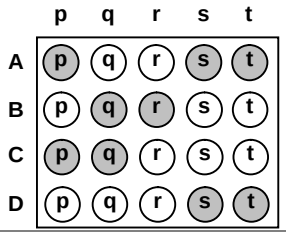
Space for rough work

SECTION – B

(Matrix-Match Type)

This section contains **2 questions**. Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s, t) in column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.

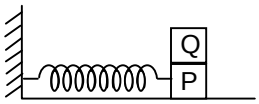
If the correct match are A-p, A-s, B-q, B-r, C-p, C-q and D-s, then the correctly bubbled 4 × 4 matrix should be as follows:



1. Relate the standard reversible process for an ideal gas in Column I to some facts in Column II corresponding to a process.

Column I		Column II	
(A)	Isochoric	(p)	Temperature remains constant.
(B)	Isobaric	(q)	Internal energy of the gas does not change.
(C)	Isothermal	(r)	Work done by the gas may be positive, negative or zero.
(D)	Adiabatic	(s)	Energy exchange between the system and surrounding takes place only through work.
		(t)	The density of the gas does not change.

2. Match the following :
A block P of mass m placed on smooth horizontal surface and connected with a spring. Another block Q of same mass is placed on the block P. The two blocks are pulled by a distance A and released. The blocks oscillates without slipping on each other. Then match the matrix.



Column – I	Column – II
(A) Friction force on the block Q, When both block coming towards equilibrium position form left. (exclude the equilibrium position)	(p) towards right
(B) Friction force at the block Q when both block going right from equilibrium position. (exclude the equilibrium position)	(q) towards left
(C) Friction force will be maximum on Q.	(r) at equilibrium position
(D) Friction force on P when both block is coming towards the equilibrium position from right. (exclude the equilibrium position)	(s) At extreme position
	(t) none of these

Space for rough work

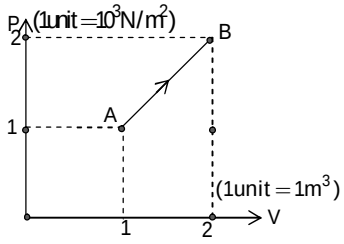
SECTION – C

Integer Answer Type

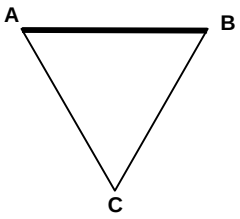
This section contains **5 questions**. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown

X	Y	Z	W
0	**0**	0	0
1	1	1	1
2	2	2	**2**
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	**9**	9

1. For P-V diagram shown for a diatomic ideal gas, heat given to the gas from A to B is(in kJ).



2. Three rods of equal length are joined to from an equilateral Δ ABC. The temperature coefficient of linear expansion is α_1 for AB, α_2 for AC and BC. For $\alpha_1 = k\alpha_2$, $\angle$ABC remains constant for small changes in temperature. Find K.



Space for rough work

3. A force $\vec{F} = \left(-y\hat{i} + x\hat{j}\right)N$ acts on a particle which undergoes counterclockwise circular motion in x-y plane in a circle of radius 4 m and with the centre at the origin. The work done by the force when the particle undergoes one complete revolution is (assume x, y are in m) $4\pi K$ Joule. Find K.

4. A body of mass m = 4 kg starts moving with velocity v_0 in a straight line is such a way that on the body work is being done at the rate which is proportional to the square of velocity as given by $P = \beta v^2$ where $\beta = \frac{0.693}{2}$. Find the time elapsed in seconds before velocity of body is doubled.

5. A sound source of frequency f_0 = 130 Hz is dropped from a height slightly greater than 250 m above the ground. At the same time a detector is thrown upwards with velocity u = 50 ms^{-1} along the same line. If the speed of sound is v = 300 ms^{-1}, if the frequency (in Hz) detected by the detector after t = 5 s is 39 × n, find the value of n (Take g = 10 ms^{-2})

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. $SiCl_4$ hydrolyses completely, its hydrolysis constant may be expressed in term of ionic product of water K_w and K_b of silicon hydroxide as

(A) $\frac{K_w}{K_b}$ (B) $\frac{K_w^2}{K_b}$

(C) $\frac{K_w^4}{K_b}$ (D) $\frac{K_w}{K_b^4}$

2. Which of the following are not acceptable sets of quantum numbers for an electron in an atom?

(1) $n=3,\ \ell=0,\ m=1,\ s=-\frac{1}{2}$ (2) $n=3,\ \ell=1,\ m=-1,\ s=\frac{1}{2}$

(3) $n=3,\ \ell=2,\ m=0,\ s=-\frac{1}{2}$ (4) $n=3,\ \ell=1,\ m=2,\ s=\frac{1}{2}$

Select the correct answer from the code given below:

(A) 1 and 2 (B) 1 and 3

(C) 1 and 4 (D) 2 and 3

3. Equilibrium constant for the given reaction is $K = 10^{20}$ at temperature 300 K:

$$\underset{(s)}{A} + \underset{(aq)}{2B} \rightleftharpoons \underset{(s)}{2C} + \underset{(aq)}{D}$$

The equilibrium concentration of B starting with mixture of 1 mole of A and 1/3 mole/litre of B at 300 K is (volume of solution of A is 1 litre)

(A) 4×10^{-11} (B) 2×10^{-10}

(C) 2×10^{-11} (D) 10^{-10}

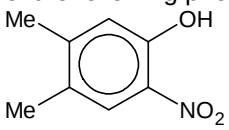
Space for rough work

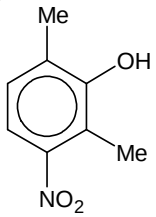
4. Three gases A, B and C are at the same pressure and temperature. If their root mean square velocities are in the ratio of $1:\frac{1}{\sqrt{2}}:\frac{1}{\sqrt{3}}$ then their vapour densities will be in the ratio

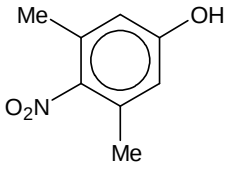
(A) 1 : 2 : 3 (B) 3 : 2 : 1

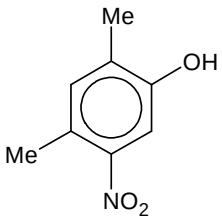
(C) $1:\sqrt{2}:\sqrt{3}$ (D) $\sqrt{3}:\sqrt{2}:\sqrt{1}$

5. Which of the following phenols will show highest acidity?

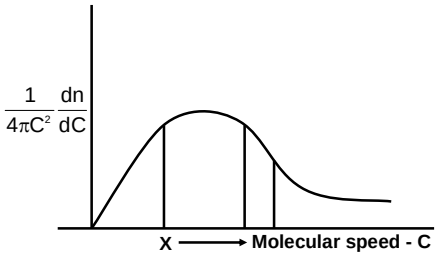
(A) 

(B) 

(C) 

(D) 

6. The Maxwell–Boltzmann distribution law of molecular speeds is graphically represented as:



The curve has which of the following characteristics:

(1) It has symmetrical distribution.
(2) The point X on axis represents the most probable speed.
(3) The area under the curve gives the total number of molecules.
(4) The maximum shift to the right as the temperature increases.

Choose the correct answer using the codes given below:

(A) 1, 2 and 3 (B) 2, 3 and 4

(C) 1 and 2 (D) 3 and 4

Space for rough work

Multiple Correct Choice Type

This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. Which of the following disproportionate (s) on heating with NaOH?
   (A) P_4 (B) S_8
   (C) Cl_2 (D) B

8. If 100 ml of 1 M H_2SO_4 is mixed with 100 ml of 98% (w/w) H_2SO_4 (d = 0.1 g/ml) then
   (A) concentration of the solution remain same
   (B) volume of the solution becomes 200 ml
   (C) mass of H_2SO_4 becomes 98 g
   (D) mass of H_2SO_4 in the solution is 19.6 g

9. In which of the following pair the first one is the stronger base than second?
   (A) CH_3COO^-, $HCOO^-$ (B) OH^-, $\bar{N}H_2$
   (C) PhOH, PhO^- (D) CH_3NH_2, CH_3OH

Paragraph Type

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple** choice questions and based on the other paragraph **3 multiple** choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 to 11

10 gm mixture of $FeSO_4$ and FeC_2O_4 on reaction with $KMnO_4$ is acidic medium utilises 100 ml of 5 M oxidising agent. When $FeSO_4$ is alone treated it used 25 ml of 0.05 M of same oxidising agent.

10. Total number of moles of ferrous ions oxidized by 1 mole of $KMnO_4$
    (A) 5/3 (B) 4/5
    (C) 2/5 (D) 5/2

11. Percentage of pure FeC_2O_4 in the mixture by weight is
    (A) 95 (B) 9.5
    (C) 90.5 (D) 25

Space for rough work

Paragraph for Question Nos. 12 to 14

Systematic arrangement of element on the basis of increasing atomic number give periodic table. This table is based on certain periodic properties which is helpful in study of the inorganic compound.

12. Select the correct arrangement:

(A) Melting point
$AgCl > KCl$

(B) Ionic radii
$Li^+ > Na^+$

(C) Ionization energy
$Be > B$

(D) Electronegativity
$Cl > F$

13. Incorrect statement among the following is
(A) thermal stability increase in case of alkaline earth metal carbonate
(B) aqueous solution of $Ca_2B_6O_{11}$ can act as a buffer in aqueous solution
(C) solubility of alkaline earth metal sulphate increase down the group
(D) ionic character of alkaline earth metal chloride increase down the group

14. Which of the following is the strongest reducing agent in aqueous solution?
(A) Na
(B) K
(C) Cs
(D) Li

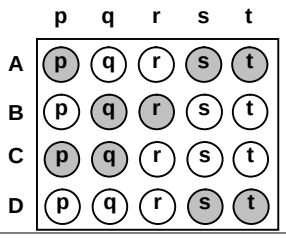
Space for rough work

SECTION – B

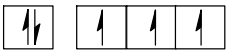
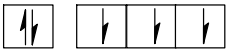
(Matrix-Match Type)

This section contains **2 questions**. Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s, t) in column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.

If the correct match are A-p, A-s, B-q, B-r, C-p, C-q and D-s, then the correctly bubbled 4 × 4 matrix should be as follows:



1. Match the following:

Column – I		Column – II	
(A)	Pauli's exclusion principle	(p)	Energy: $\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s$
(B)	Hund's rule of maximum multiplicity	(q)	For B_2 $(\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 \mid (\pi 2p_y)^{m_s=-\frac{1}{2}} (\pi 2p_z)^{m=-\frac{1}{2}} \mid$
(C)	Aufbau energy rule	(r)	For Z = 29 $[Ar]\ 4s^1 3d^{10}$
(D)	Stability of half filled orbitals	(s)	For Z = 7 [He]  [He] 
		(t)	For Z = 6 $[He]\, 2s^2 (2p_x)^{m_s=+\frac{1}{2}} (2p_y)^{m_s=+\frac{1}{2}}$

Note: M_s = magnetic spin quantum number.

2. Match the column – I with chemicals of that in valence in the process:

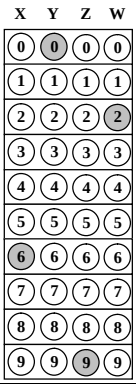
Column – I		Column – II	
(A)	Nelson cell	(p)	NaOH
(B)	Down's process	(q)	NaCl (aqueous)
(C)	Castner's process	(r)	Cl_2
(D)	Castner's kellner process	(s)	H_2
		(t)	NaCl (molten)

Space for rough work

SECTION – C

Integer Answer Type

This section contains **5 questions**. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown



1. The equivalent weight for underline molecule in the following reaction is M/n (M = molar mass of P_4). Find n.

   $\underline{P_4} \longrightarrow PH_3 + H_2PO_2^-$

2. Number of Si – O – Si bond in $[Si_3O_9]^{-6}$ is/are.

3. The number of compounds / species in which underlined atoms are sp^2 hybridized will be.

   $\underline{C}O_2, \underline{B}O_2^-, \underline{B}O_3^{3-}, H_3\underline{B}O_3, \underline{B}F_3, H\underline{C}HO, CH_3\underline{C}OOH, \underline{C}O_3^{2-}, \bar{\underline{C}}H_3, \underline{C}(\text{graphite}), \underline{C}(\text{diamond})$

4. In an experiment one mole of $Na_2CO_3.10H_2O$ was heated to a temperature by which it got converted into $Na_2CO_3xH_2O$. Water vapour obtained in this process were used to convert $CuSO_4$ into $CuSO_4.5H_2O$. If 0.6 moles of $CuSO_4$ was hydrated, value of x is.

5. At 300 K and one atmospheric presure 2 L of N_2O_4 (g) was heated to 600 K at constant presure. 50 % of N_2O_4 (g) got dissociated according to the reaction:

   $N_2O_4(g) \rightleftharpoons N_2(g) + 2O_2(g)$

   Final volume of the mixture in litres is.

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **6 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. If, $\cos^6\alpha + \sin^6\alpha + k(\sin 2\alpha)^2 = 1, \forall\ \alpha \in \left(0, \frac{\pi}{2}\right)$, then k is

   (A) $\frac{3}{4}$ (B) $\frac{1}{4}$

   (C) $\frac{1}{3}$ (D) $\frac{1}{8}$.

2. In a ΔABC, if $\angle A = 30^\circ$ and $BC = 2 + \sqrt{5}$, then the distance of the vertex A from the orthocentre of the triangle is

   (A) 1 (B) $\sqrt{3}(2+\sqrt{5})$

   (C) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (D) $\frac{1}{2}$.

3. The remainder when $(27)^{40}$ is divided by 12 is

   (A) 3 (B) 7

   (C) 9 (D) 11.

4. If in a ΔABC, tanA + tanB + tanC > 0, then the triangle is

   (A) Acute-angled (B) obtuse-angled

   (C) right-angled (D) none of these.

Space for rough work

5. In a ΔABC, if $\cos A + \cos B + \cos C = \frac{7}{4}$, then, $\frac{R}{r}$ is equal to

(A) $\frac{3}{4}$ (B) $\frac{4}{3}$

(C) $\frac{2}{3}$ (D) $\frac{3}{2}$.

6. In a ΔABC, if $(a + b + c)(b + c - a) = \lambda bc$, then

(A) $\lambda < 0$ (B) $\lambda > 6$

(C) $0 < \lambda < 4$ (D) $\lambda > 4$.

Multiple Correct Choice Type

This section contains **3 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

7. Which of following functions have the maximum value unity?

(A) $\sin^2 x - \cos^2 x$ (B) $\sqrt{\frac{6}{5}}\left(\frac{1}{\sqrt{2}}\sin x + \frac{1}{\sqrt{3}}\cos x\right)$

(C) $\cos^6 x + \sin^6 x$ (D) $\cos^2 x + \sin^4 x$.

8. If, $\sin\theta + \cos\theta = -\frac{1}{5}$, then a value of $\tan\frac{\theta}{2}$ satisfies which of the following equations?

(A) $2x^2 - x - 1 = 0$ (B) $x^2 - 2x - 3 = 0$

(C) $2x^2 - 5x - 3 = 0$ (D) $2x^3 - 7x^2 + 2x + 3 = 0$.

9. The sides of a ΔABC satisfy the equation $2a^2 + 4b^2 + c^2 = 4ab + 2ac$, then.

(A) the triangle is isosceles (B) the triangle is obtuse

(C) $B = \cos^{-1}\left(\frac{7}{8}\right)$ (D) $A = \cos^{-1}\left(\frac{1}{4}\right)$

Space for rough work

Paragraph Type

This section contains **2 paragraphs.** Based upon one of paragraphs **2 multiple** choice questions and based on the other paragraph **3 multiple** choice questions have to be answered. Each of these questions has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

Paragraph for Question Nos. 10 to 11

Read the following write up carefully and answer the following questions:

Let a_n denote the number of all n-digit positive integers formed by the digits 0, 1 or both such that no consecutive digits in them are 0. Let b_n = the number of such n-digit integers ending with digit 1 and c_n = the number of such n-digit integers ending with digit 0.

10. The value of b_6 is
(A) 7 (B) 8
(C) 9 (D) 11.

11. Which of the following is correct?
(A) $a_{17} = a_{16} + a_{15}$ (B) $a_{17} \neq a_{16} + a_{15}$
(C) $a_{17} \neq b_{16} + c_{16}$ (D) $a_{17} = c_{17} + c_{16}$.

Paragraph for Question Nos. 12 to 14

Read the following write up carefully and answer the following questions:

Let a, b, c are the sides opposite to angles A, B, C respectively in a ΔABC and $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\frac{c}{2}$ and $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$, If a = 6, b = 3 and $\cos(A - B) = \frac{4}{5}$

12. Angle C is equal to
(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$
(C) $\frac{3\pi}{4}$ (D) $\frac{2\pi}{3}$.

13. Area of the triangle is equal to
(A) 8 (B) 9
(C) 10 (D) 11.

14. Value of sinA is equal to
(A) $\frac{1}{\sqrt{5}}$ (B) $\frac{2}{\sqrt{5}}$
(C) $\frac{1}{2\sqrt{5}}$ (D) $\frac{1}{\sqrt{3}}$

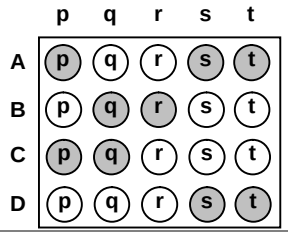
Space for rough work

SECTION – B

(Matrix-Match Type)

This section contains **2 questions**. Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s, t) in column II. The answers to these questions have to be appropriately bubbled as illustrated in the following example.

If the correct match are A-p, A-s, B-q, B-r, C-p, C-q and D-s, then the correctly bubbled 4×4 matrix should be as follows:



1. Match the following Column-I with Column-II

Column – I		Column – II	
(A)	The total number of selections of atleast one of the fruits which can be made from 3 bananas, 4 apples, and 2 oranges is	(p)	Greater than 50
(B)	If 7 points out of 12 are in the same straight line, then the number of triangles formed is	(q)	Greater than 100
(C)	The number of ways of selecting 10 balls from unlimited number of red, black, white and green balls is	(r)	Greater than 150
(D)	The total number of proper divisors of 38808 is	(s)	Greater than 200
		(t)	Greater than 300

2. Match the following Column-I with Column-II

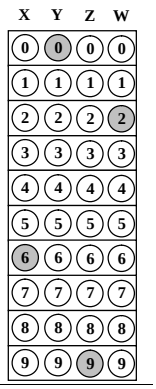
Column – I		Column – II	
(A)	If, $\cos A = \frac{\sin B}{2\sin C}$, then ΔABC is	(p)	isosceles
(B)	If, $\frac{\cos A + 2\cos C}{\cos A + 2\cos B} = \frac{\sin B}{\sin C}$, then ΔABC may be	(q)	obtuse angle
(C)	If, $\frac{2\cos A}{a} + \frac{\cos B}{b} + \frac{2\cos C}{c} = \frac{a}{bc} + \frac{b}{ca}$, then Δ is	(r)	right angle
(D)	If, $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A - B)}{\sin(A + B)}$, then ΔABC may be	(s)	acute angle
		(t)	equilateral

Space for rough work

SECTION – C

Integer Answer Type

This section contains **5 questions**. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown



1. Let $a_1, a_2, a_3, \ldots a_{100}$ be on arithmetic progression with $a_1 = 3$ and $s_p = \sum_{i=1}^{p} a_i, 1 \le p \le 100$. For any integer n with $1 \le n \le 20$, let m = 5n. If $\frac{S_m}{S_n}$ does not depend on n, then a_2 is _____

2. A pack contains n cards numbered from 1 to n. Two consecutive cards are removed from the pack and the sum of the numbers on the remaining cards is 1224. If the smaller of the numbers on the removed cards is k. Then k – 20 = _____

3. If z is any complex number satisfying $|z - 3 - 2i| \le 2$, then the minimum value of $|2z - 6 + 5i|$ is _____.

4. If a, b, c are integers, not all simultaneously equal and ω is cube root of unity ($\omega \ne 1$), then minimum value of $|a + b\omega + c\omega^2|$ is _____

5. Consider a triangle ABC and let a, b, and c denote the lengths of the sides opposite the vertices A, B and C respectively. Suppose a = 6, b = 10 and the area of the triangle is $15\sqrt{3}$, if $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to _____

Space for rough work